Quantitative Trading: Theory and Practice
Advanced Course – Lecture 2

Multi-Factor Combination and Portfolio Construction

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Lecture 2 Positioning

- In Lecture 1, we studied how to move from raw data to factor construction and factor testing.
- In this lecture, we take the next step: how to turn factor information into a tradable portfolio.
- The central workflow is

$$\text{multiple factors} \to \text{combined signal} \to \text{candidate stock pool} \to \text{weights} \to \text{strategy}.$$

- The goal is not only to rank assets, but to decide what to hold, how much to hold, and what type of strategy to build.

Learning Objectives

By the end of this lecture, students should be able to:

- explain why multiple factors are often combined in practice;
- understand several common methods for multi-factor combination;
- distinguish between a signal, a candidate stock pool, and a final portfolio;
- construct portfolio weights from a combined factor signal;
- understand the difference between long-only, long–short, and alpha strategies;
- interpret portfolio construction choices in terms of diversification, turnover, and implementation.

Why Single-Factor Testing Is Not Enough

A single factor may be useful, but it is rarely the whole story.

- different factors capture different sources of information;
- one factor may work well in one regime and poorly in another;
- a single factor can be noisy or unstable;
- combining multiple factors can improve robustness and reduce dependence on one style.

Main idea

Factor research ends with a score. Portfolio construction begins with decisions.

From Multiple Factors to a Combined Signal

Suppose we have K factors:

$$F_{t,i}^{(1)},\ F_{t,i}^{(2)},\ \ldots,\ F_{t,i}^{(K)}.$$

Our goal is to transform them into a single composite score:

$$S_{t,i} = \Psi\left(F_{t,i}^{(1)}, \ldots, F_{t,i}^{(K)}\right).$$

- $S_{t,i}$ summarizes the overall attractiveness of asset i at time t;
- the combined signal is the basis for stock selection and weight construction;
- therefore, signal combination is the bridge from factor research to portfolio formation.

Method 1: Equal-Weight Combination

The simplest approach is to average standardized factor values:

$$S_{t,i} = \frac{1}{K} \sum_{k=1}^{K} F_{t,i}^{(k)}.$$

- easy to implement and explain;
- treats all factors symmetrically;
- useful as a baseline model.

Advantage: simplicity and transparency.
Limitation: it ignores the possibility that some factors are more informative or more stable than others.

Method 2: Weighted Combination

A more flexible approach is to assign different weights to different factors:

$$S_{t,i} = \sum_{k=1}^{K} \omega_k F_{t,i}^{(k)}, \qquad \sum_{k=1}^{K} \omega_k = 1.$$

Possible principles for choosing ω_k include:

- prior economic judgement;
- historical IC or RankIC performance;
- robustness or stability across subperiods.

Key point: weighted combination allows the model to express confidence in some factors more than others.

Method 3: Rank Aggregation

Instead of combining raw scores directly, we may first rank each factor cross-sectionally and then average the ranks:

$$S_{t,i} = \frac{1}{K} \sum_{k=1}^{K} \operatorname{rank}\left(F_{t,i}^{(k)}\right).$$

- more robust to outliers and scale differences;
- often easier to interpret in cross-sectional stock selection;
- naturally connects with RankIC analysis from Lecture 1.

This is a popular and practical way to build a composite score.

Why Standardization Matters Before Combination

Before combining factors, their scales should usually be made comparable.
Common preprocessing steps include:

- winsorization to reduce the effect of extreme values;
- z-score standardization;
- rank transformation;
- neutralization when sector or size effects should be controlled.

Reason

Without preprocessing, one factor may dominate the composite score simply because its numerical scale is larger.

From Combined Signal to Candidate Stock Pool

The combined signal is not yet a portfolio. It is first used to define a candidate stock pool.

- the signal provides an ordering of assets;
- the candidate pool is a subset of assets chosen from that ordering;
- the final portfolio is obtained only after assigning weights.

Conceptual distinction

$$\text{signal} \neq \text{candidate pool} \neq \text{final weighted portfolio.}$$

Common Rules for Candidate Stock Selection

Three common methods are:

Top-N selection

- choose the N highest-scoring assets.

Quantile selection

- choose assets in the top quantile, for example the top 20%.

Threshold selection

- choose all assets whose score exceeds a given threshold.

The choice depends on the desired portfolio size, diversification, and trading style.

Top-N Selection

Top-N selection chooses a fixed number of assets with the highest factor scores:

$$\mathcal{C}_t = \{i : \ \mathrm{rank}(s_{t,i}) \leq N\},$$

where $s_{t,i}$ is the combined score of asset i at time t.

- The portfolio size is fixed and easy to control.
- It is suitable when we want a stable number of holdings.
- It is commonly used in long-only stock selection strategies.

Practical implication

If $N = 50$, the strategy always selects 50 stocks, regardless of whether the overall market signal is strong or weak.

Pros and Cons of Top-N Selection

Advantages

- Simple and intuitive.
- Portfolio size is stable.
- Easy to combine with equal-weight or risk-based weighting.

Disadvantages

- The cutoff score changes over time.
- Some selected stocks may have weak signals when the whole universe has low scores.
- It ignores the absolute strength of the factor signal.

Key issue

Top-N controls the number of assets, but not the quality of the selected signals.

How Many Assets Should We Choose?

A natural question in Top-N selection is:

How large should N be?

From portfolio theory, total risk can be decomposed into:

$$\text{Total Risk} = \text{Systematic Risk} + \text{Unsystematic Risk}.$$

- **Systematic risk** is market-wide risk and cannot be eliminated by diversification.
- **Unsystematic risk** is firm-specific risk and can be reduced by holding more stocks.
- As more stocks are added, total portfolio risk decreases, but at a decreasing rate.

Practical implication

In a simple Top-N stock selection strategy, choosing $N \approx 30$ is a common starting point: it provides basic diversification without overly diluting the factor signal.

Diversification Benefit Diminishes with N

For an equally weighted portfolio, assume that the stock-specific risks are roughly independent. Then the unsystematic risk component decreases as the number of holdings increases:

$$\text{Unsystematic Variance} \propto \frac{1}{N}, \qquad \text{Unsystematic Volatility} \propto \frac{1}{\sqrt{N}}.$$

Number of Assets N	$1/\sqrt{N}$	Diversification Effect
1	1.00	No diversification
5	0.45	Large risk reduction
10	0.32	Further risk reduction
30	0.18	Unsystematic risk is substantially reduced
100	0.10	Additional benefit becomes smaller

The first few stocks bring the largest diversification benefit. After around 30 stocks, adding more names still helps, but the marginal benefit becomes much smaller.

Cutoff-Based Selection

Both quantile selection and threshold selection can be viewed as **cutoff-based selection**.

$$\mathcal{C}_t = \{i: \ s_{t,i} \geq c_t\},$$

where $s_{t,i}$ is the combined score of asset i at time t, and c_t is a cutoff.

- **Quantile selection:** the cutoff is determined by the cross-sectional distribution.

$$c_t = q_{1-\alpha,t}.$$

- **Threshold selection:** the cutoff is fixed or pre-specified.

$$c_t = c.$$

Key idea

Both methods select assets by comparing their scores with a cutoff. The main difference is how the cutoff is chosen.

Dynamic Cutoff vs. Fixed Cutoff

Cutoff-based selection has two common forms.

Dynamic cutoff: quantile selection

- Example: select the top 20% of assets.
- The cutoff changes over time with the score distribution.
- The selected proportion is stable.

Fixed cutoff: threshold selection

- Example: select all assets with score larger than 0.5.
- The cutoff is fixed across time.
- The selected number depends on how strong the signals are.

Important distinction

Quantile selection controls the **relative rank**, while fixed threshold selection controls the **absolute signal strength**.

Pros and Cons of Cutoff-Based Selection

Advantages

- Flexible and easy to implement.
- Directly links candidate selection to factor scores.
- Useful when the size of the tradable universe changes over time.

Potential issues

- A quantile cutoff may select weak signals when the whole market has low scores.
- A fixed threshold may select too few or too many assets in different periods.
- The final portfolio size may be unstable.

Practical implication

Cutoff-based rules are often combined with Top-N selection. For example, select assets above the top 20% cutoff, then keep at most 30 names.

Combining Different Selection Rules

In practice, candidate stock selection rarely relies on only one rule.
Many index products and thematic ETFs combine several rules, such as:

- **universe filtering**: define the initial eligible universe;
- **risk or quality filtering**: remove unsuitable stocks;
- **cutoff-based selection**: keep stocks satisfying some factor criteria;
- **Top-N selection**: choose a fixed number of final constituents.

Key message

Top-N, cutoff-based selection, and risk filtering are often used together, but answer different questions:

$$\text{eligible?} \quad \rightarrow \quad \text{good factor exposure?} \quad \rightarrow \quad \text{how many assets?}$$

Example: CSI 300 Dividend Low Volatility Index

The CSI 300 Dividend Low Volatility Index is as an example of combining several rules.

Index objective

Select 50 securities from the CSI 300 universe that have:

- continuous dividend payments,
- high dividend yields,
- low volatility.

- **Initial universe**: CSI 300 constituents.
- **Final number**: 50 securities.
- **Weighting method**: dividend-yield weighting.

Interpretation

This is not a pure Top-N strategy. It first filters the universe, then applies factor-based selection, and finally fixes the number of index constituents.

Example: Selection Procedure

The index construction procedure can be summarized as follows.

1. **Universe definition**
    - Start from CSI 300 constituents.
2. **Eligibility and risk filtering**
    - Require continuous cash dividends in the past three years.
    - Require the average dividend payout ratio to be between 0 and 1.
    - Keep stocks whose ROE volatility ranks in the lowest 80%.
3. **Cutoff-based factor selection**
    - Rank remaining stocks by average cash dividend yield.
    - Keep the top 50% as the candidate pool.
4. **Top-N final selection**
    - Rank candidates by one-year volatility in ascending order.
    - Select the lowest-volatility 50 stocks.

Link: https://www.csindex.com.cn/#/indices/family/detail?indexCode=930740

Risk Filtering Before Portfolio Formation

Before constructing the candidate stock pool, we may exclude assets with unusually high implementation or governance risk.

Typical filters include:

- special treatment (ST or *ST) stocks;
- stocks under trading suspension or with abnormal trading status;
- newly listed stocks with insufficient trading history;
- extremely illiquid assets;
- assets with missing or unreliable data.

Main idea

A strong factor signal does not necessarily imply that the asset is suitable for trading. Portfolio construction should start from an investable universe rather than the full raw universe.

Why Filter Again Before Portfolio Formation?

In the previous factor testing stage, we already filtered the stock universe. However, this does not mean that the filtering effect will automatically remain valid throughout the whole pipeline.

A practical data-processing issue

In factor testing, filtered stocks are often handled by setting their factor exposures to `NaN`. But during factor combination, these missing values may be transformed into zeros after standardization, alignment, filling, or score aggregation.

A filtered stock may have exposure `NaN` during factor testing. After factor combination, this `NaN` may become exposure 0. Some valid stocks may have negative combined factor exposures. Therefore, a previously filtered stock with exposure 0 may rank above valid stocks with negative exposures.

Potential problem

A stock that should not be tradable may re-enter the candidate pool simply because its missing exposure has been converted into a neutral score.

Two Roles of Filtering

The two filtering steps have different roles in the trading pipeline.

- **Filtering in factor testing:** makes the factor evaluation cleaner by removing abnormal or unreliable observations.
- **Filtering before portfolio formation:** guarantees that the final candidate pool only contains tradable and eligible assets.
- **Reason for repeated filtering:** intermediate data processing may change `NaN` exposures into numerical scores, allowing previously excluded stocks to reappear in the ranking.

Key message

Factor testing asks whether the signal works.
Portfolio filtering asks whether the selected assets are really eligible for trading.

Why Exclude Special-Treatment Stocks?

Special-treatment stocks often carry risks that are not well described by ordinary factor signals.

- they may face financial distress or severe operating uncertainty;
- their prices can be strongly driven by event risk rather than normal cross-sectional patterns;
- liquidity and execution quality may be poor;
- they may distort portfolio risk and weaken the stability of backtest results.

Interpretation

Filtering out ST or other high-risk stocks is not a prediction step. It is a universe-control step designed to improve tradability and risk consistency.

Why Exclude Firms with Fraud Risk?

Besides special-treatment stocks, firms with elevated financial fraud risk should also be excluded from the investable universe.

- accounting manipulation can make reported fundamentals unreliable;
- factor signals based on earnings, value, or profitability may become misleading;
- fraud risk often comes with sudden price crashes, trading disruptions, and severe reputational events;
- such firms may create backtest profits that are not stable or economically credible.

Main idea

If the accounting information is unreliable, then the corresponding factor signal is unreliable as well. Therefore, fraud-risk filtering is a natural part of universe control.

Example Indicators of Financial Fraud Risk

Possible warning indicators include:

- unusually rapid growth in accounts receivable relative to revenue;
- persistent divergence between net profit and operating cash flow;
- abnormally high accruals;
- frequent downward earnings revisions or restatements;
- unusually large related-party transactions;
- sharp increases in inventories without matching sales growth;
- weak interest coverage despite reported profit growth;
- repeated qualified, adverse, or disclaimer audit opinions.

Interpretation

These indicators do not prove fraud by themselves, but they can be used as red flags to identify firms requiring exclusion or closer scrutiny.

Filtered Universe and Candidate Stock Pool

Let $\mathcal{U}_t$ denote the full stock universe at time t, and let $\mathcal{F}_t$ denote the set of assets removed by risk filters. Then the investable universe becomes

$$\mathcal{U}_t^{\mathrm{inv}} = \mathcal{U}_t \setminus \mathcal{F}_t.$$

The candidate stock pool is selected only from the filtered universe:

$$\mathcal{C}_t \subseteq \mathcal{U}_t^{\mathrm{inv}}.$$

- first apply risk and tradability filters;
- then rank the remaining assets by the combined signal;
- finally construct portfolio weights on the selected candidate set.

Key point: filtering should be done before ranking and weighting.

Selection with Retention and Risk Control

After excluding high-risk assets, we still do not want to rebuild the portfolio completely at every rebalance date.

- some previously selected assets may still have sufficiently strong composite signals;
- retaining part of the existing holdings can reduce unnecessary turnover;
- this helps avoid excessive trading caused by small changes in rank;
- the retained assets must still satisfy the current risk filters and tradability requirements.

Idea

At each rebalance date, keep some previously held assets if they remain investable and still rank near the top of the current signal ordering.

This gives a more stable candidate stock pool and connects stock selection with both turnover control and risk management.

Why Do We Need Retention?

If we always select the current top-ranked stocks, the portfolio may change too frequently.

- A stock may move from rank 30 to rank 32 only because of a small score change.
- If the rule strictly selects top 30 stocks, this stock will be removed immediately.
- Another stock may enter the portfolio simply because its rank improves slightly.
- Such small rank changes may cause unnecessary trading.

Problem

A pure ranking-based selection rule may generate excessive turnover, even when the underlying signal changes only slightly.

Key idea

We do not need to replace a holding immediately as long as it still has a reasonably good signal and satisfies all current risk and tradability filters.

Retention with a Buffer Zone

A common solution is to use a **buffer zone** around the selection cutoff.

Suppose the target number of holdings is $N = 30$.

- **New stocks:** select new stocks only if they rank in the top 30.
- **Existing holdings:** retain them if they still rank within the top 50.
- **Risk filters:** retained stocks must still pass current risk and tradability filters.

$$\text{Buy threshold} = 30, \qquad \text{Retention threshold} = 50.$$

Interpretation

The top 30 rule controls the target portfolio size, while the top 50 retention rule avoids unnecessary replacement of existing holdings.

Important

Retention is not unconditional. A stock should not be retained if it becomes suspended, high-risk, illiquid, or fails other eligibility requirements.

Retention, Turnover, and Risk Control

Retention connects candidate selection with portfolio implementation.

- **Signal perspective:** keep stocks whose signals are still reasonably strong.
- **Trading-cost perspective:** reduce unnecessary buying and selling caused by small ranking fluctuations.
- **Risk-control perspective:** remove stocks that fail current filters, even if they were held previously.
- **Portfolio-stability perspective:** make the candidate pool more persistent across rebalance dates.

Key message

Retention is a compromise between signal chasing and portfolio stability.

Implementation principle

First check whether the stock is still investable; then decide whether its signal is good enough to be retained.

What Is a Rebalance Date?

A **rebalance date** is the date when we update the portfolio.
At each rebalance date, we usually:

- update the tradable universe;
- recompute factor exposures and composite scores;
- apply risk filters and candidate selection rules;
- decide which existing holdings to retain;
- adjust portfolio weights.

Key idea

A trading strategy is not rebuilt every day unless we explicitly choose daily rebalancing. The rebalance date defines when the strategy is allowed to update its holdings.

Common Choices of Rebalance Frequency

The rebalance frequency determines how often the portfolio is updated.

Frequency	Example	Typical Feature
Daily	every trading day	very responsive, high turnover
Weekly	every Friday	medium responsiveness
Monthly	month-end or month-beginning	common in equity factor strategies
Quarterly	quarter-end	lower turnover, slower response
Semi-annually	June and December	common in index construction

Practical note

For many medium- to low-frequency stock selection strategies, monthly rebalancing is a common starting point.

How Should We Choose Rebalance Dates?

The choice of rebalance frequency should match the nature of the signal.

- **Fast-moving signals**
  - examples: short-term reversal, short-term momentum, trading volume;
  - may require daily or weekly rebalancing.
- **Slow-moving signals**
  - examples: value, profitability, dividend yield, quality;
  - usually suitable for monthly, quarterly, or semi-annual rebalancing.
- **Data update frequency**
  - the portfolio should not rebalance more frequently than the signal is meaningfully updated.

Important principle

A higher rebalance frequency does not necessarily mean a better strategy. It may only create more turnover and transaction costs.

Rebalance Frequency and Trading Costs

More frequent rebalancing allows the strategy to react faster, but it also increases trading costs.

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot \text{turnover}_t.$$

- Frequent rebalancing may capture new signals more quickly.
- But it usually increases turnover.
- Higher turnover leads to higher transaction costs.
- After costs, the net performance may be much weaker than the gross backtest result.

Trade-off

Rebalance frequency should balance signal freshness and implementation cost.

A Practical Rule for Rebalance Date Selection

A practical way to choose rebalance dates is to ask three questions.

1. **How often does the signal change?**
   - If the factor changes slowly, frequent rebalancing is unnecessary.
2. **How expensive is trading?**
   - If transaction costs are high, rebalance less frequently.
3. **How stable is the selected portfolio?**
   - If rankings fluctuate heavily, use retention or a lower rebalance frequency.

For our course project

We can start with monthly rebalancing, then compare it with weekly or quarterly rebalancing in backtesting.

Extension: What Is High-Frequency Trading?

High-frequency trading (HFT) refers to trading strategies that operate at very short time scales, often from milliseconds to seconds.

- **Data:** tick-level data, order book data, transaction data.
- **Signals:** short-lived patterns in prices, quotes, order flow, and liquidity.
- **Execution:** fast order submission, cancellation, and position adjustment.
- **Holding period:** very short, usually much shorter than daily or monthly factor strategies.

Key distinction

High-frequency trading is not simply "daily rebalancing with more speed". It uses different data, different signals, and different execution infrastructure.

Why HFT Is Different from Factor Strategy

Our factor strategy is a low- to medium-frequency portfolio selection framework. HFT is different in several important ways.

Aspect	Factor Strategy	High-Frequency Trading
Time scale	days to months	milliseconds to seconds
Data	daily / monthly data	tick and order book data
Main task	stock selection	order placement and execution
Cost focus	transaction cost and turnover	spread, slippage, latency, market impact
Infrastructure	backtesting pipeline	low-latency trading system

Practical message

Increasing rebalance frequency does not automatically turn a factor strategy into HFT. When the frequency becomes very high, execution quality and market microstructure become central issues.

HFT and Technology Infrastructure

In HFT, trading performance depends not only on the strategy model, but also on data speed, communication networks, server location, and execution technology.

- **Low-latency network**
  - Orders and market data must be transmitted with very small delays.
  - Even millisecond-level differences may affect trading outcomes.
- **Co-location**
  - Trading servers are often placed close to exchange matching engines.
  - This reduces the physical transmission distance and improves execution speed.
- **Market data advantage**
  - Some firms try to process order book updates faster than competitors.
  - Faster reaction may create an advantage in order placement and cancellation.
- **Order anticipation and fairness concerns**
  - In some controversial cases, traders may infer or anticipate other participants' orders and trade ahead of them.
  - Such behavior raises concerns about market fairness, transparency, and regulation.

HFT and Market Fairness Concerns

High-frequency trading may raise market fairness concerns when speed advantages are used to exploit slower market participants.

- **Front-running-like behavior**
  - Some traders may detect or infer large incoming orders.
  - They may trade ahead of these orders to profit from the expected price movement.
  - This creates a potential unfair advantage over slower investors.
- **Scalping-like behavior**
  - Traders may use very short-term price changes to repeatedly capture tiny profits.
  - In a problematic form, this may involve exploiting privileged information, faster access, or induced price movements.
- **Spoofing**
  - Traders place large orders without genuine trading intention.
  - These orders may create a false impression of supply or demand.
  - After other traders react, the fake orders are cancelled.

The key issue is not high speed itself, but whether the trading behavior creates false market signals, exploits unfair information advantages, or harms market integrity.

From Candidate Assets to Portfolio Weights

Once the filtered candidate stock pool is determined, the next step is to assign portfolio weights.

General form

For each date t, construct weights $w_{t,i}$ on the selected set such that

$$\sum_i w_{t,i} = 1$$

for a fully invested long-only portfolio, or according to the exposure constraints of the chosen strategy.

The weighting rule determines:

- how much capital is allocated to each selected asset;
- how concentrated or diversified the portfolio becomes;
- how strongly the portfolio tilts toward the highest-ranked signals;
- how sensitive the portfolio is to turnover and implementation costs.

Weighting Scheme 1: Equal Weight

For a candidate set $\mathcal{C}_t$ with N assets,

$$w_{t,i} = \begin{cases} \frac{1}{N}, & i \in \mathcal{C}_t, \\ 0, & \text{otherwise.} \end{cases}$$

- easy to implement;
- naturally diversified;
- avoids overreacting to small score differences.

Limitation: it does not distinguish between moderately attractive and highly attractive assets.

Weighting Scheme 2: Score-Based Weighting

A more signal-sensitive approach is to assign larger weights to assets with stronger combined scores.

One simple idea is:

$$w_{t,i} \propto S_{t,i}, \qquad i \in \mathcal{C}_t,$$

with normalization chosen to satisfy the exposure constraint.

- stronger signals receive more capital;
- the portfolio becomes more closely tied to the ranking strength;
- but concentration risk may increase.

Weighting Scheme 3: Market-Capitalization Weighting

For each selected asset $i \in \mathcal{C}_t$, define its market capitalization as

$$\mathrm{MktCap}_{t,i} = P_{t,i} \times \mathrm{Shares}_{t,i}.$$

Then the portfolio weight is given by

$$w_{t,i} = \frac{\mathrm{MktCap}_{t,i}}{\sum_{j \in \mathcal{C}_t} \mathrm{MktCap}_{t,j}}.$$

- larger companies receive larger portfolio weights;
- this method is widely used in index construction;
- weights naturally adjust with stock prices;
- therefore, it usually requires less frequent rebalancing than equal weighting.

Practical interpretation

Market-cap weighting is not designed to maximize factor exposure. Instead, it provides a simple, scalable, and low-turnover weighting scheme.

Weighting Scheme 4: Risk-Aware Weighting

Portfolio weights can also incorporate a risk measure such as volatility.
A basic example is inverse-volatility weighting:

$$w_{t,i} \propto \frac{1}{\sigma_{t,i}},$$

possibly combined with a factor score.

- reduces the influence of highly volatile assets;
- introduces basic risk awareness into portfolio formation;
- often leads to more balanced contributions across holdings.

This is a useful step before moving to more formal optimization-based methods.

Portfolio Constraints in Practice

Real portfolios are built under constraints.
Common constraints include:

- **long-only:** $w_{t,i} \geq 0$;
- **fully invested:** $\sum_i w_{t,i} = 1$;
- **position cap:** $w_{t,i} \leq w_{\max}$;
- **limited number of holdings:** control the portfolio size;
- **turnover control:** avoid excessive trading from one period to the next.

Key point: portfolio construction is not only about expected return; it is a constrained decision problem.

Weight Rebalancing

After portfolio weights are determined, market prices keep changing. Therefore, the actual portfolio weights may drift away from the target weights. Let

$$w_{t,i}^{\text{target}}$$

be the desired weight at rebalance date t, and let

$$w_{t,i}^{\text{current}}$$

be the current weight before trading.
A full rebalance would trade the difference:

$$\Delta w_{t,i} = w_{t,i}^{\text{target}} - w_{t,i}^{\text{current}}.$$

- If we always trade to exactly match target weights, turnover may be high.
- Small weight differences may not be worth trading.
- In practice, we may allow small deviations from target weights.
- Weight rebalancing should balance target exposure and trading cost.

Threshold-Based Weight Rebalancing

To reduce unnecessary turnover, we can rebalance only when the weight deviation is large enough. Define the weight deviation:

$$d_{t,i} = \left| w_{t,i}^{\text{target}} - w_{t,i}^{\text{current}} \right|.$$

A simple threshold rule is:

$$\text{trade asset } i \quad \Longleftrightarrow \quad d_{t,i} > \tau,$$

where τ is a pre-specified rebalance threshold.

- If $d_{t,i} \leq \tau$, keep the current position unchanged.
- If $d_{t,i} > \tau$, trade toward the target weight.
- A larger τ reduces turnover but allows larger tracking error.
- A smaller τ tracks target weights more closely but increases trading.

Trade-off

Threshold-based rebalancing reduces unnecessary trades, but the actual portfolio may deviate from the desired factor exposure.

Strategy Type 1: Long-Only Portfolio

A long-only strategy buys assets with strong signals and does not short weak assets.

- most realistic for many institutional settings;
- directly relevant in markets with short-sale restrictions;
- focuses on absolute return and benchmark-relative performance.

A typical construction is:

- select the top-ranked assets;
- assign equal weights or score-based weights;
- rebalance periodically.

Strategy Type 2: Long–Short Portfolio

A long–short strategy buys high-score assets and shorts low-score assets.

- emphasizes relative pricing and cross-sectional alpha;
- reduces dependence on overall market direction;
- closely related to grouped return spreads from Lecture 1.

A basic long–short construction is:

$$w_{t,i} = \begin{cases} +\dfrac{1}{|L_t|}, & i \in L_t, \\ -\dfrac{1}{|S_t|}, & i \in S_t, \\ 0, & \text{otherwise}, \end{cases}$$

where L_t is the long set and S_t is the short set.

Strategy Type 3: Alpha Portfolio

An alpha strategy aims to earn excess return after controlling for broad market exposure. Possible interpretations include:

- a benchmark-relative long-only portfolio;
- a market-neutral or approximately beta-neutral portfolio;
- a strategy designed to isolate stock selection skill from general market movement.

Simple benchmark-relative idea

$$r_t^{\alpha} = r_t^{\text{portfolio}} - r_t^{\text{benchmark}}.$$

The main objective is not just to make money, but to generate excess performance beyond broad market exposure.

Comparing the Three Strategy Types

	Long-only	Long–short	Alpha
Main objective	absolute return	relative-value spread	excess return beyond benchmark or market exposure
Exposure to market direction	high	lower	controlled
Use of weak signals	ignored	shorted	underweighted or hedged
Interpretation	practical investing	factor premium test	stock selection skill

Rebalancing and Turnover Control

Portfolio construction is dynamic, not static.

- factor scores change through time;
- candidate sets and weights must be updated;
- but frequent rebalancing increases turnover and trading cost.

Typical ways to manage turnover include:

- lower rebalance frequency;
- retain part of the previous holdings;
- smooth portfolio weights;
- require a stronger signal change before trading.

From Weights to Portfolio Performance

After the portfolio is formed, we evaluate its realized performance.

Basic period return

$$\mathrm{PnL}_t = w_{t-1}^\top r_t.$$

NAV recursion

$$\mathrm{NAV}_t = \mathrm{NAV}_{t-1}(1 + \mathrm{PnL}_t).$$

At this stage, portfolio design decisions can be assessed through metrics such as return, volatility, drawdown, Sharpe ratio, and turnover.

Methodological Summary of Today

The methodology of this lecture can be summarized as follows:

1. combine multiple factors into one composite signal;
2. transform the signal into a candidate stock pool;
3. assign portfolio weights under practical constraints;
4. choose a strategy type: long-only, long–short, or alpha;
5. manage turnover through rebalancing design;
6. evaluate cumulative portfolio performance.

Coding Lab for This Lecture

In the programming session, students will:

1. construct a composite signal from several factors;
2. generate a candidate stock pool using top-N or top-quantile selection;
3. compare equal-weight and score-based weighting schemes;
4. build a long-only portfolio;
5. build a simple long–short portfolio;
6. compute strategy NAV and compare performance across strategy types.

Takeaways

- Multi-factor combination is a practical way to improve robustness and diversify information sources.
- A composite signal must still be transformed into a candidate stock pool and a weighted portfolio.
- Long-only, long–short, and alpha strategies use the same signal in different ways.
- Portfolio construction choices strongly affect concentration, turnover, and implementation quality.
- Next lecture: trading costs, backtesting discipline, and robustness checks.

Thank you!